

Merrivoss Bioanalysis Inc, a Tarnovell company

OB11 GPCR antagonists: five-isoform CYP inhibition at screening concentration

Final Report

Study MV-2022-176 (LIMS M22/BIO/04063)

Client: Orvessant Biosciences Pty Ltd

Study lead: Dr M. Larkin, DMPK Study Director

Report date: August 10, 2022

1 Summary

40 compounds from the Orvessant piperidyl benzamide series 11 were tested for Orvessant Biosciences Pty Ltd in the Five-isoform DDI screen. Testing started on July 25, 2022 and results were reported on August 10, 2022. All accepted runs met matrix acceptance criteria. Inhibition above 50% at 10 μ M was most frequent for CYP2C9.

Table 1 Study summary

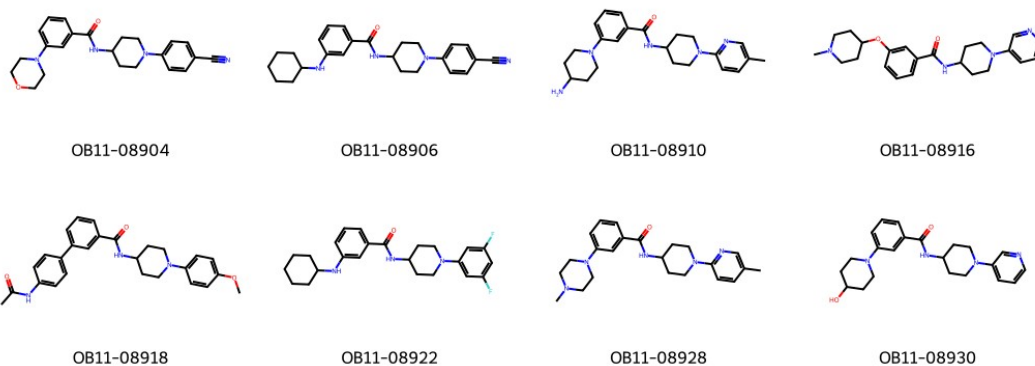
Metric	Value
Project	MV-2022-176
Client	Orvessant Biosciences Pty Ltd
Compounds tested	40
Series / chemotype	Orvessant piperidyl benzamide series 11 (Piperidyl benzamide)
Assays	CYP inhibition panel (1A2/2C9/2C19/2D6/3A4) at 10 μ M
Turnaround (calendar days)	16
Price	USD 12,400
CYP1A2 >50% inhibition @10 μ M	10
CYP2C9 >50% inhibition @10 μ M	12
CYP2C19 >50% inhibition @10 μ M	11
CYP2D6 >50% inhibition @10 μ M	11
CYP3A4 >50% inhibition @10 μ M	11
Compounds with any CYP >50%	18
Ketoconazole (3A4) IC50 (μ M)	0.031

2 Objective

To rank the test article set for Five-isoform DDI screen liability so that Orvessant Biosciences Pty Ltd could choose analogues for progression.

3 Test items

40 compounds were received as 10 mM DMSO solutions in one accession lot on July 25, 2022 and logged against M22/BIO/04063. Identity and concentration were taken as supplied by the client. Stocks were stored at -20 °C and thawed once for testing.



4 Methods

Five-isoform DDI screen

Pooled human liver microsomes (0.1 mg/mL protein) were incubated with each compound at 10 μ M and an isoform-selective probe substrate at approximately K_m , in duplicate, for 10 minutes at 37 °C after NADPH addition. Metabolite formation is measured by LC-MS/MS against vehicle (0.1% DMSO) incubations; the lower limit of quantitation of each metabolite is at least 20-fold below the vehicle signal.

Probe substrates and controls

Isoform	Probe substrate	Metabolite	Positive control
CYP1A2	Phenacetin	Acetaminophen	Furafylline
CYP2C9	Diclofenac	4'-Hydroxydiclofenac	Sulfaphenazole
CYP2C19	S-Mephenytoin	4'-Hydroxymephenytoin	Ticlopidine
CYP2D6	Dextromethorphan	Dextrorphan	Quinidine
CYP3A4	Midazolam	1'-Hydroxymidazolam	Ketoconazole

Positive controls are run as 7-point concentration responses on every run. Furafylline and ticlopidine (mechanism-based inhibitors) are preincubated with microsomes and NADPH for 15 minutes before probe addition. A run is accepted when each control IC₅₀ falls within 0.5-2 times the laboratory historical mean. Results are reported as mean percent inhibition relative to vehicle; values above 50% at 10 μ M are flagged, and small negative values indicate no inhibition.

5 Results

CYP inhibition results

Inhibition above 50% at 10 μ M was most frequent for CYP2C9. In total 18 compounds inhibited at least one isoform by more than 50% at 10 μ M (CYP3A4: 11; CYP2D6: 11; CYP2C19: 11; CYP2C9: 12;

CYP1A2: 10). All control IC50s were within 0.5-2 times the historical mean (ketoconazole 0.031 μ M against a historical mean of 0.030 μ M).

Table 3 Percent inhibition at 10 μ M

Compound	CYP1A2 (%)	CYP2C9 (%)	CYP2C19 (%)	CYP2D6 (%)	CYP3A4 (%)
OB11-08904	12	12	-9	6	2
OB11-08906	16	56	6	20	34
OB11-08910	7	4	43	3	4
OB11-08916	8	-8	11	59	37
OB11-08918	65	93	93	7	95
OB11-08922	20	54	41	96	34
OB11-08928	3	8	19	10	2
OB11-08930	4	11	1	10	-1
OB11-08934	-4	8	5	5	-7
OB11-08936	20	23	73	60	54
OB11-08942	1	13	33	1	-4
OB11-08948	1	77	70	3	56
OB11-08952	9	2	12	23	-5
OB11-08964	12	4	35	10	-3
OB11-08970	88	91	100	5	44
OB11-08972	87	79	67	21	76
OB11-08978	-0	3	13	92	-3
OB11-08982	-12	1	3	5	2
OB11-08984	-1	5	2	1	-7
OB11-08990	58	95	100	88	65
OB11-08994	-4	4	1	13	-4
OB11-08996	-5	20	7	18	-9
OB11-09000	9	3	3	21	1
OB11-09008	94	93	92	75	100
OB11-09012	2	6	23	8	-1
OB11-09026	9	2	27	0	14
OB11-09030	6	3	9	1	9
OB11-09032	96	89	94	33	97
OB11-09036	7	10	38	-4	5
OB11-09042	2	-0	1	4	-0
OB11-09044	6	-1	-2	-12	-1
OB11-09048	10	2	33	5	3
OB11-09054	59	40	9	78	66
OB11-09060	9	7	18	97	26
OB11-09062	9	8	4	75	13
OB11-09066	100	100	92	96	87
OB11-09074	6	-3	6	6	9
OB11-09080	83	96	89	45	85
OB11-09086	100	88	67	72	80
OB11-09090	6	-4	5	13	9

Single concentration 10 μ M, duplicate incubations, mean reported; negative values = no inhibition

Controls (IC50, μ M): furafylline (CYP1A2) 1.9; sulfaphenazole (CYP2C9) 0.45; ticlopidine (CYP2C19) 1.2; quinidine (CYP2D6) 0.064; ketoconazole (CYP3A4) 0.031

Table 4 Positive control IC50s

Control	Isoform	IC50 (µM)	Historical mean (µM)
Furafylline	CYP1A2	1.9	1.8
Sulfaphenazole	CYP2C9	0.45	0.35
Ticlopidine	CYP2C19	1.2	1.2
Quinidine	CYP2D6	0.064	0.060
Ketoconazole	CYP3A4	0.031	0.030

6 Deviations

No deviations from the study plan.

7 Discussion

Inhibition above 50% at 10 µM was most frequent for CYP2C9. Values are in vitro and do not by themselves predict clinical risk; free exposure estimates are needed for that assessment.

8 Conclusion

Recommended next steps: run concentration responses for compounds flagged on CYP2C9 and any other isoform above 50%.

9 Records

Raw data, assay worksheets and instrument files are retained for 5 years under M22/BIO/04063.

Name	Role	Signature	Date
Megan Larkin	DMPK Study Director	<u>M Larkin</u>	August 10, 2022
Helen Marchwood	Bioanalytical Quality Manager	<u>H Marchwood</u>	August 9, 2022